STU FCHPT

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Deep Koopman Economic Model Predictive Control of a Pasteurisation Unit

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ecc26 EUROPEAN CONTROL CONFERENCE

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Better Predictions, Better Decisions?

- -32% economic cost reduction
- Classic identification (N4SID) → Koopman model
- Same structure of control loop – Economic MPC

Five Questions to be Answered

- Why is pasteurization economically challenging?
- Why does model quality matter for EMPC?
- How to isolate the effect of model change?
- Does Koopman model actually predict better?
- Does that improvement translate into €?


1
2
3
4
5
CAUTION
ROTATING
SHAFT

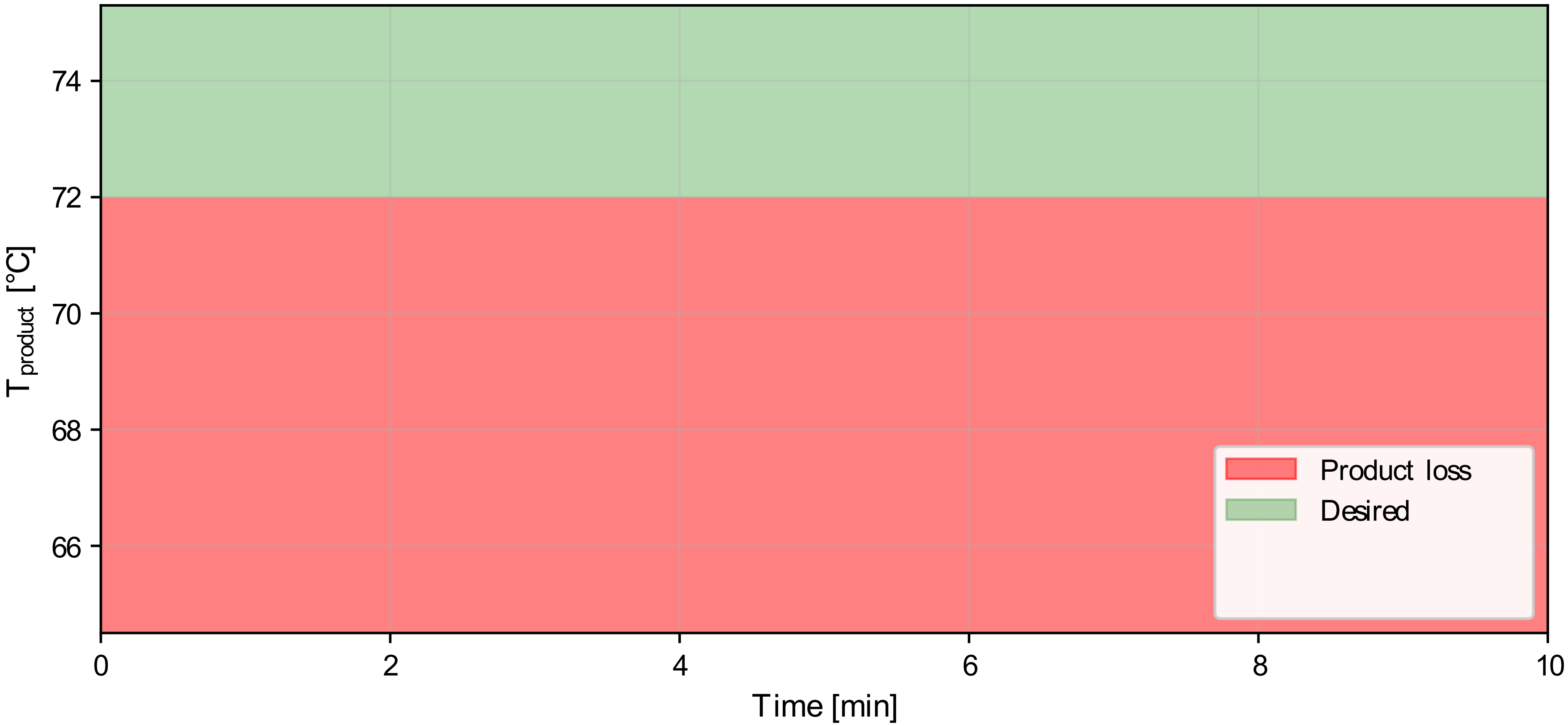
Economic Challenge of Pasteurization

- Am I making product - €?
  - Quality of the product
  - Volume of the product
  - Minimize waste
- Am I making it efficiently?
  - Minimize energy consumption
  - Ensuring Quality

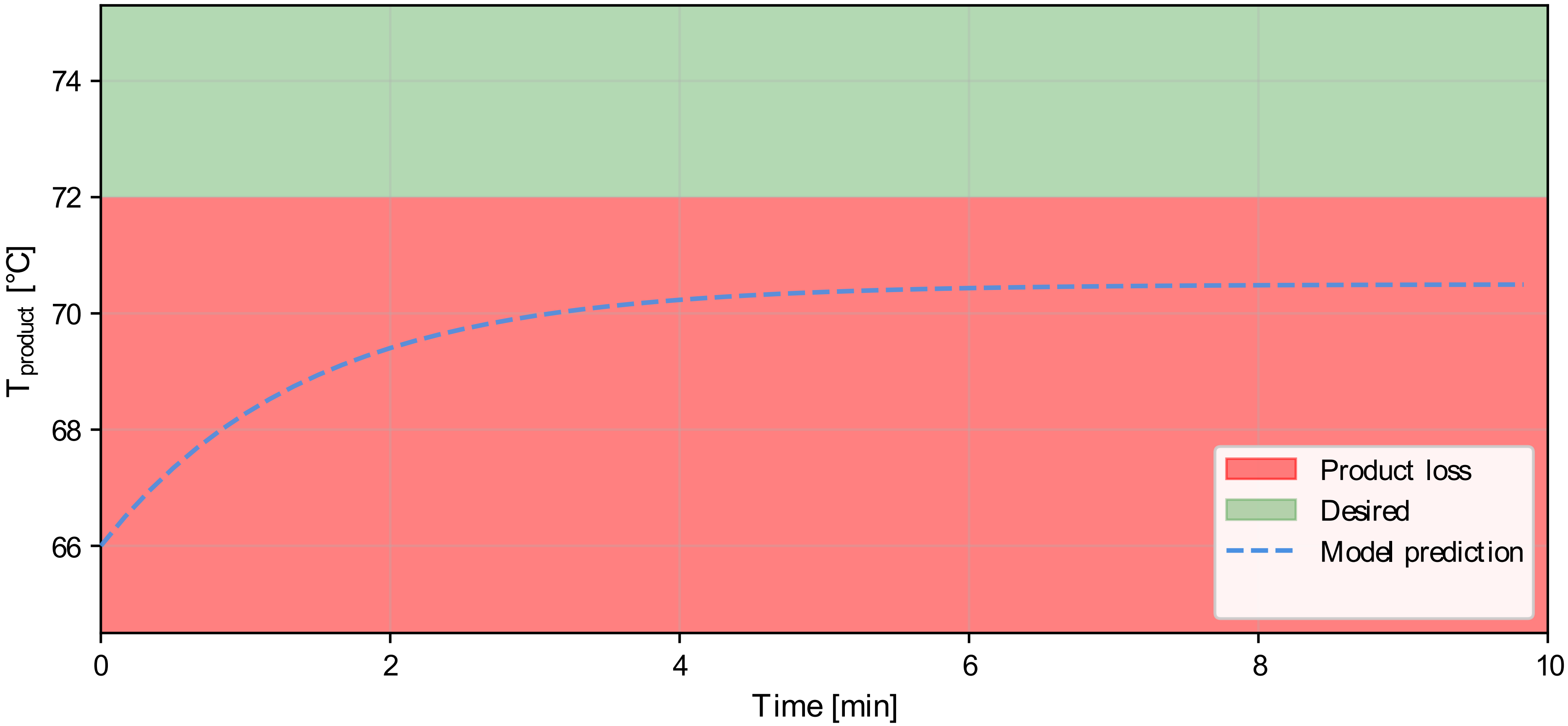
min Product loss

s.t. Quality constraints

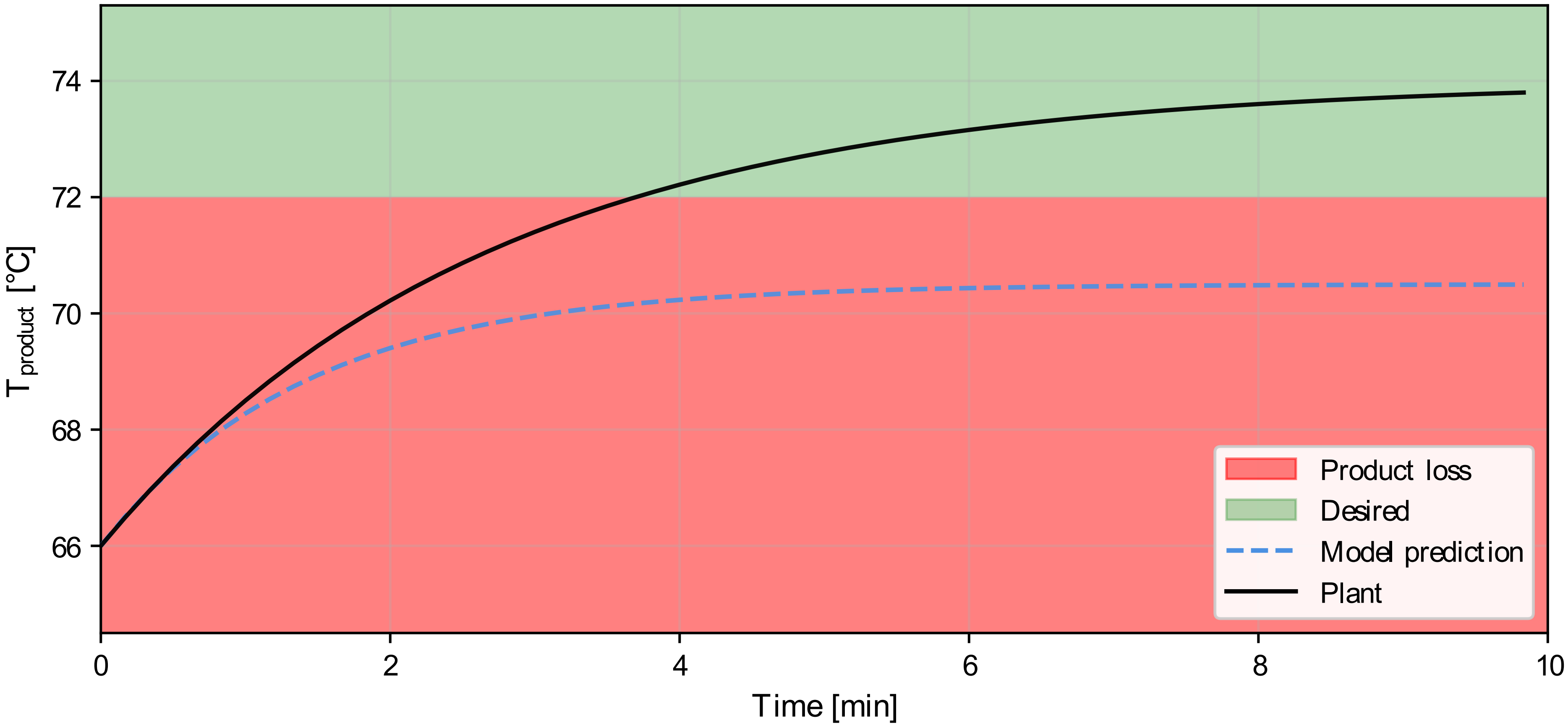
Model Quality is Economic Performance



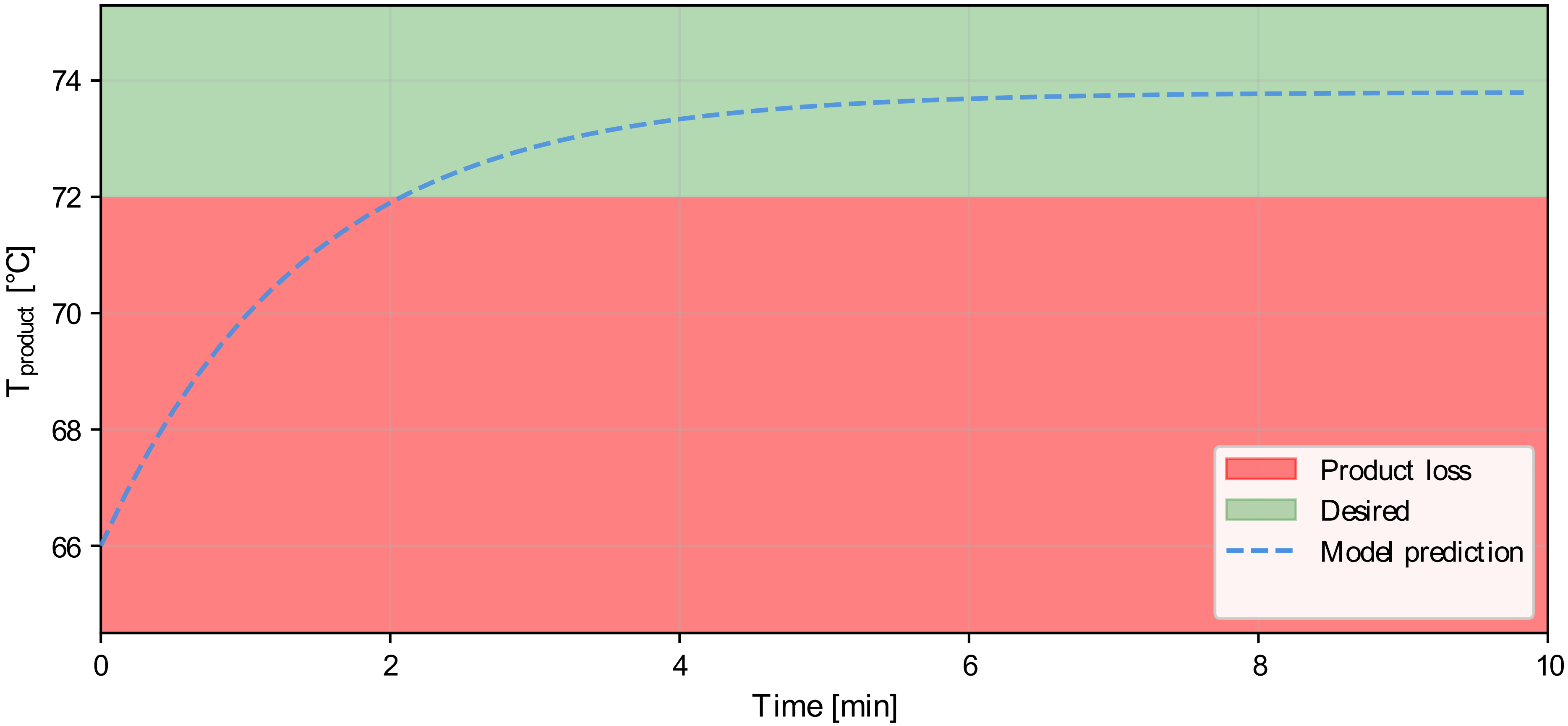
Model Quality is Economic Performance



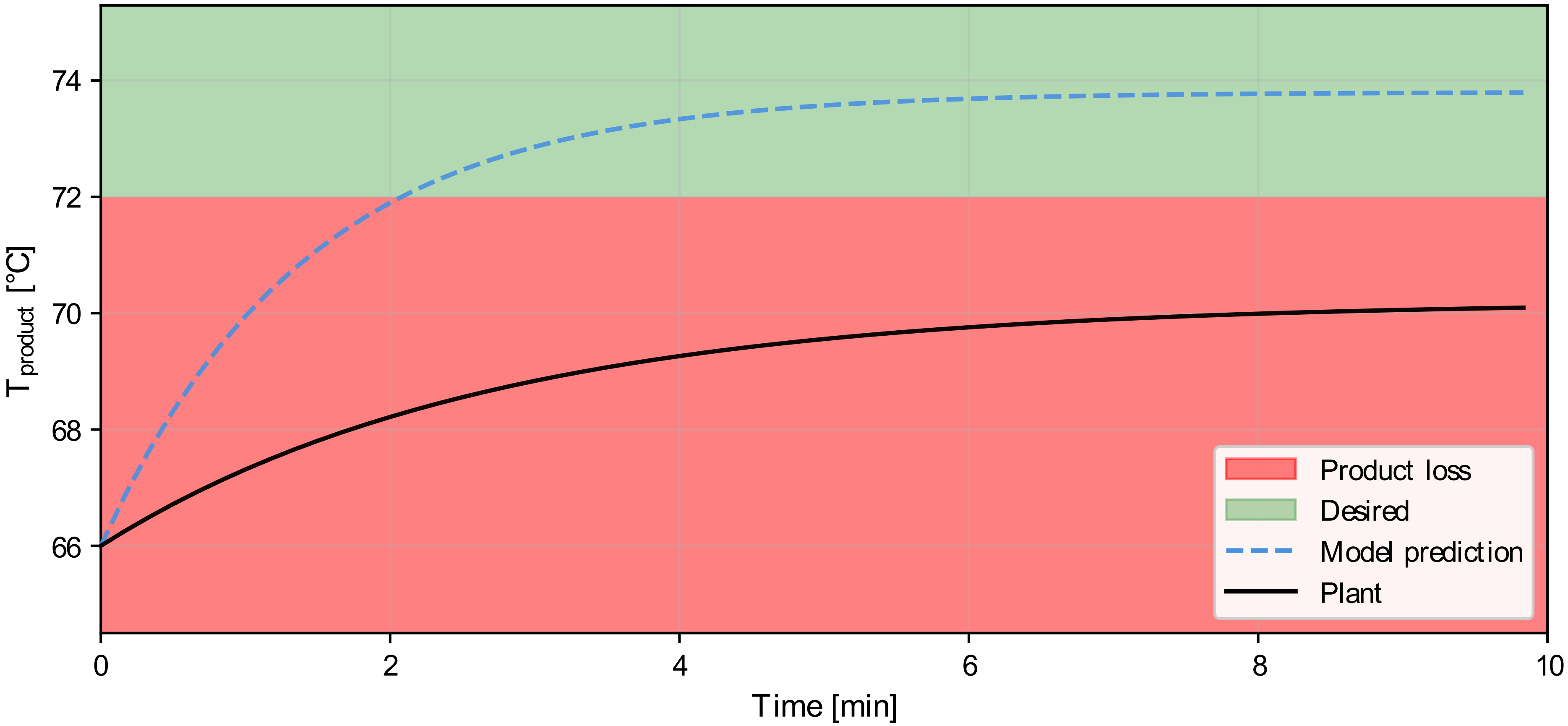
Model Quality is Economic Performance



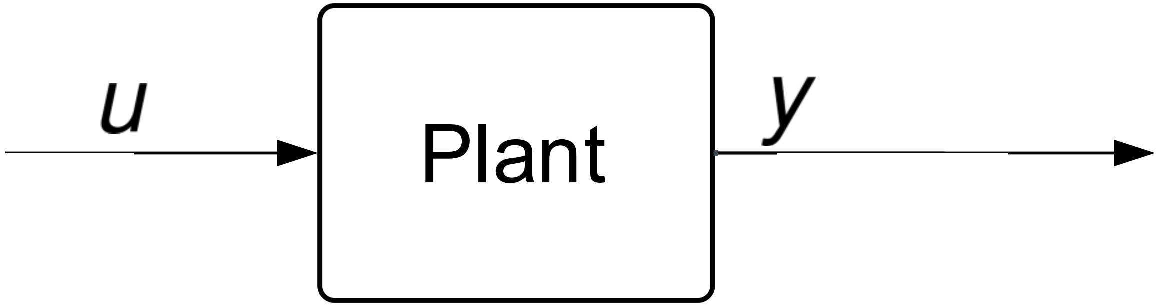
Model Quality is Economic Performance



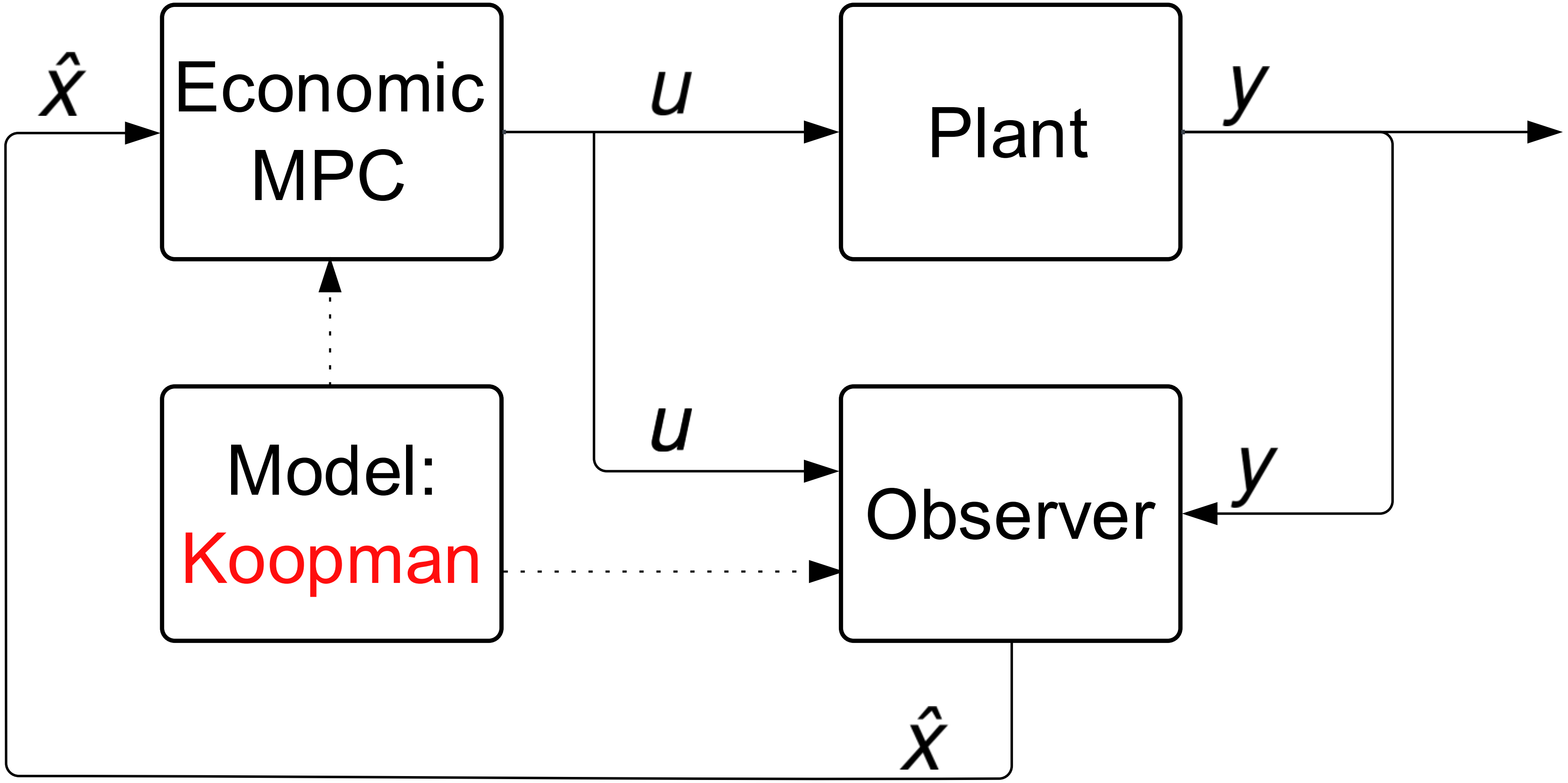
Model Quality is Economic Performance



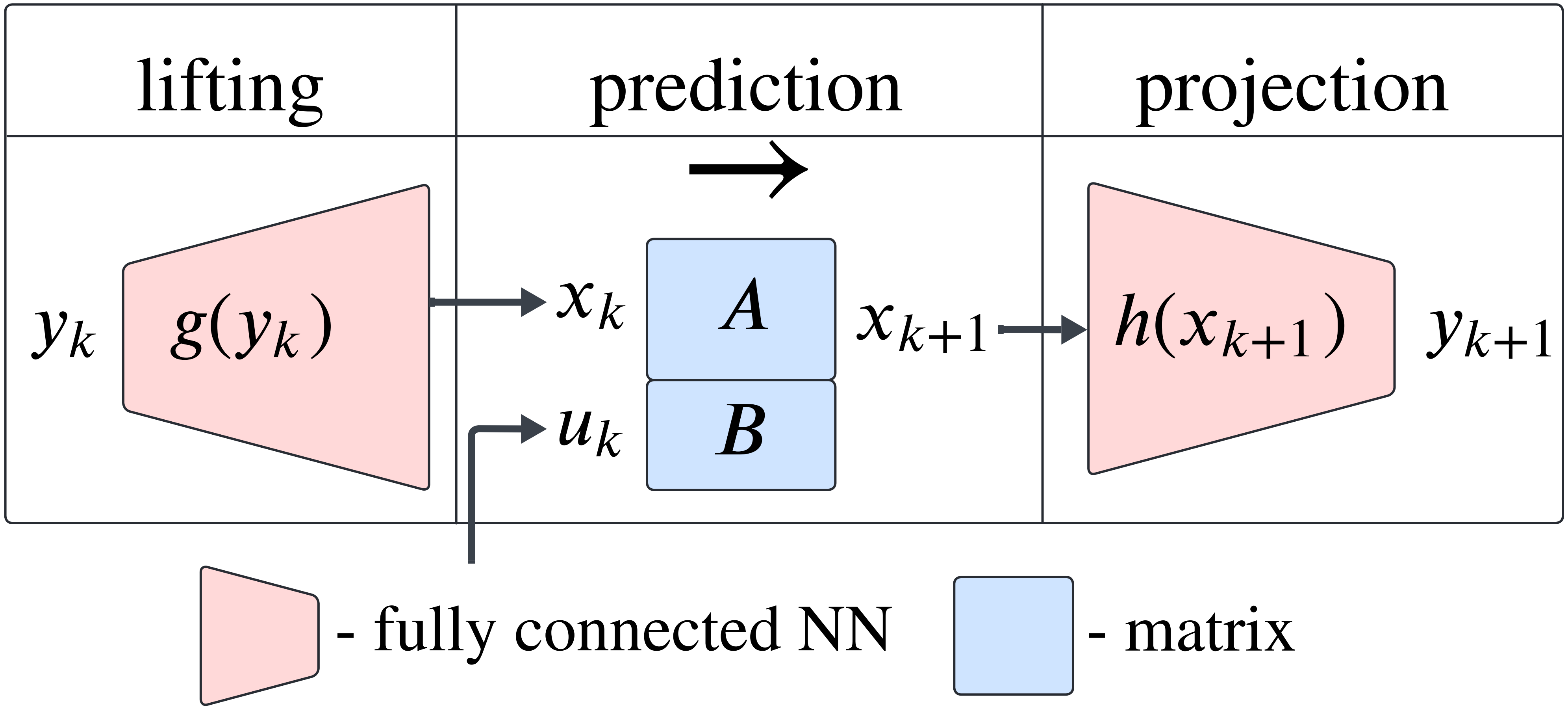
Simulation close loop



Fair Evaluation: Only Model Changes

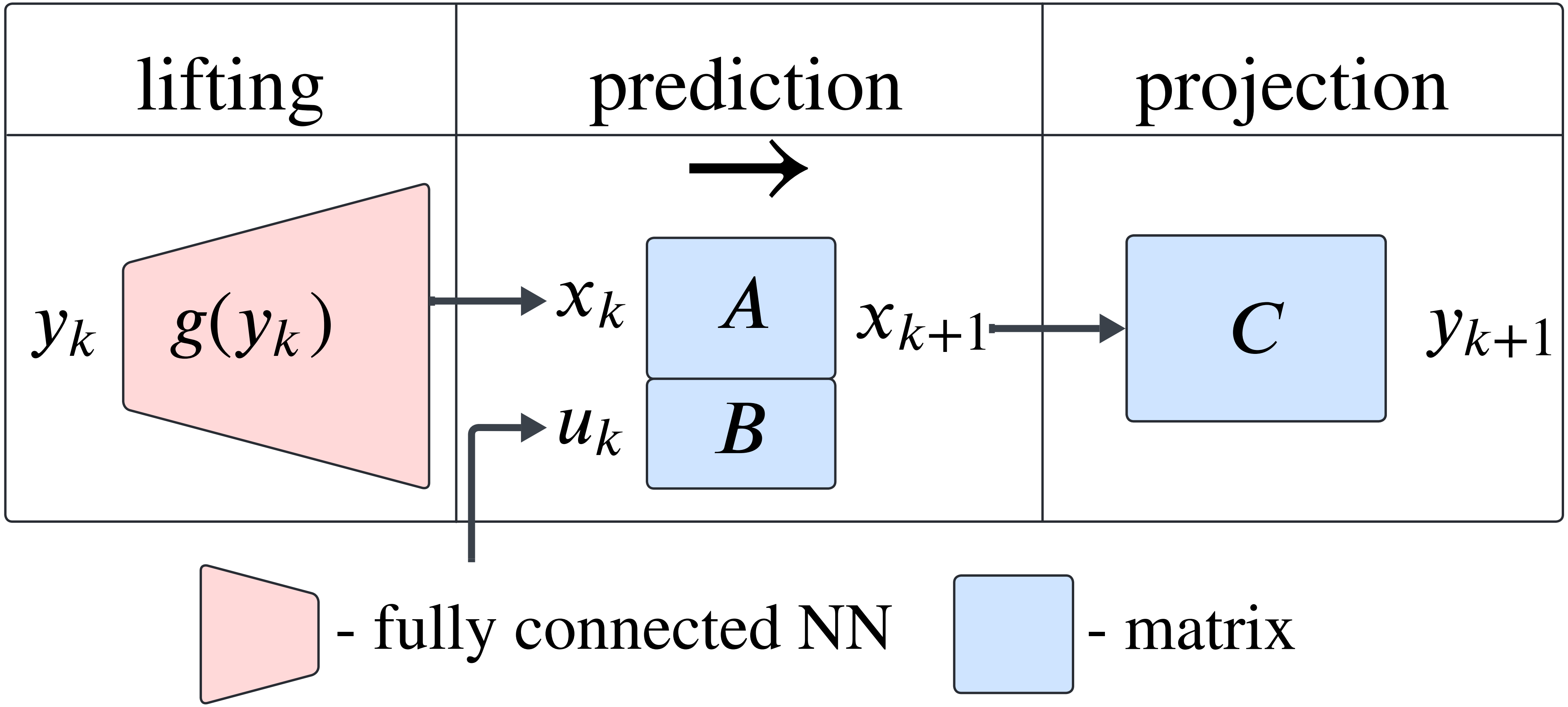


Deep Koopman

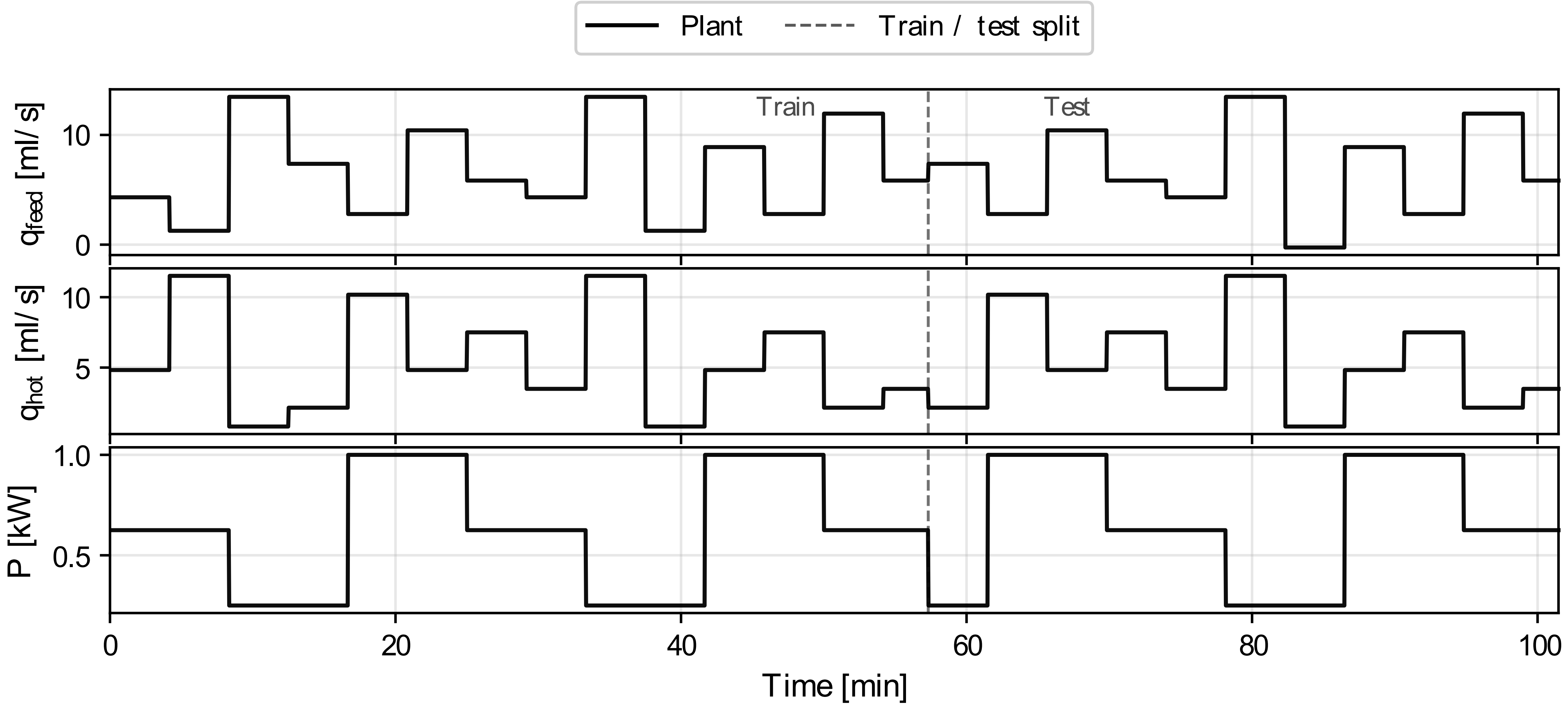


Lusch, Bethany, J. Nathan Kutz, and Steven L. Brunton. "Deep learning for universal linear embeddings of nonlinear dynamics." Nature communications 9.1 (2018)

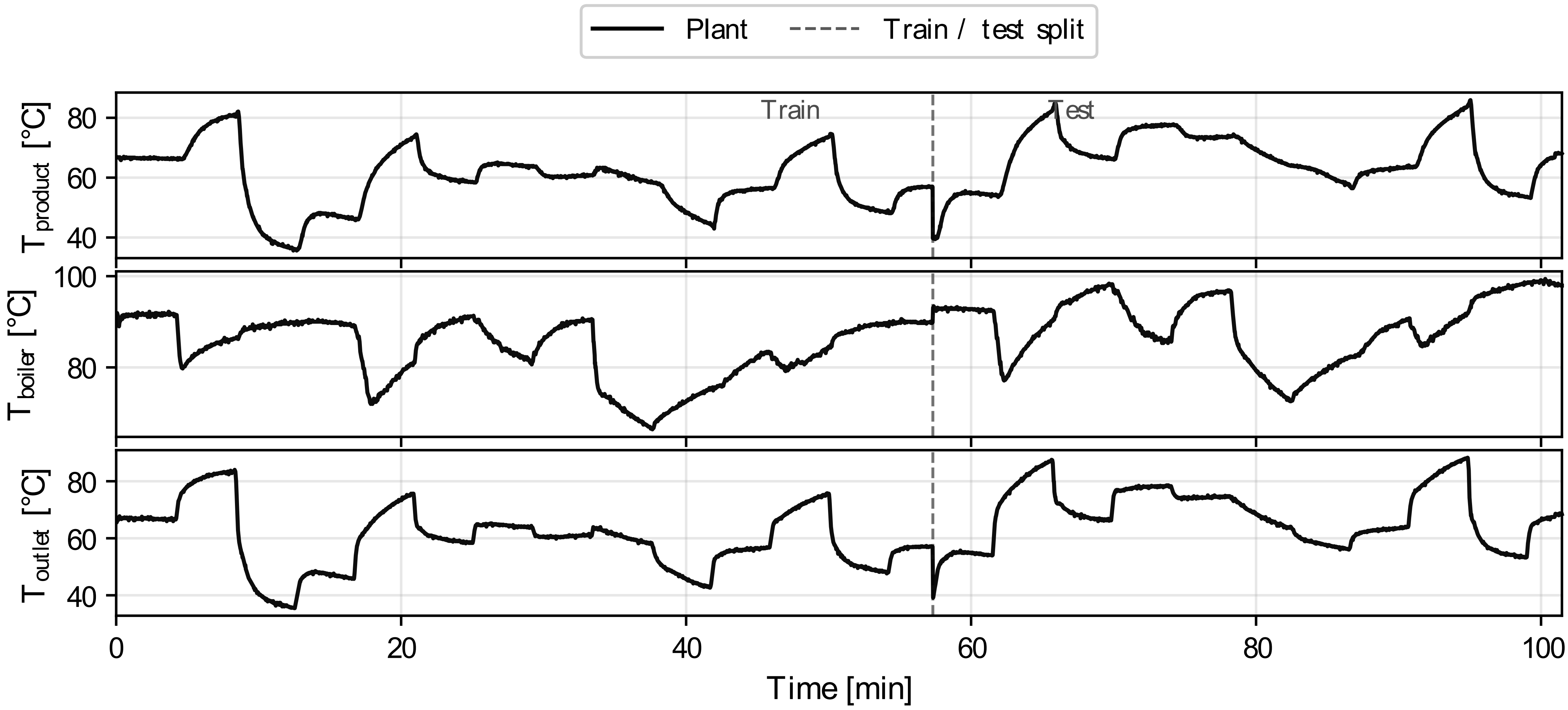
Deep Koopman – QP ready



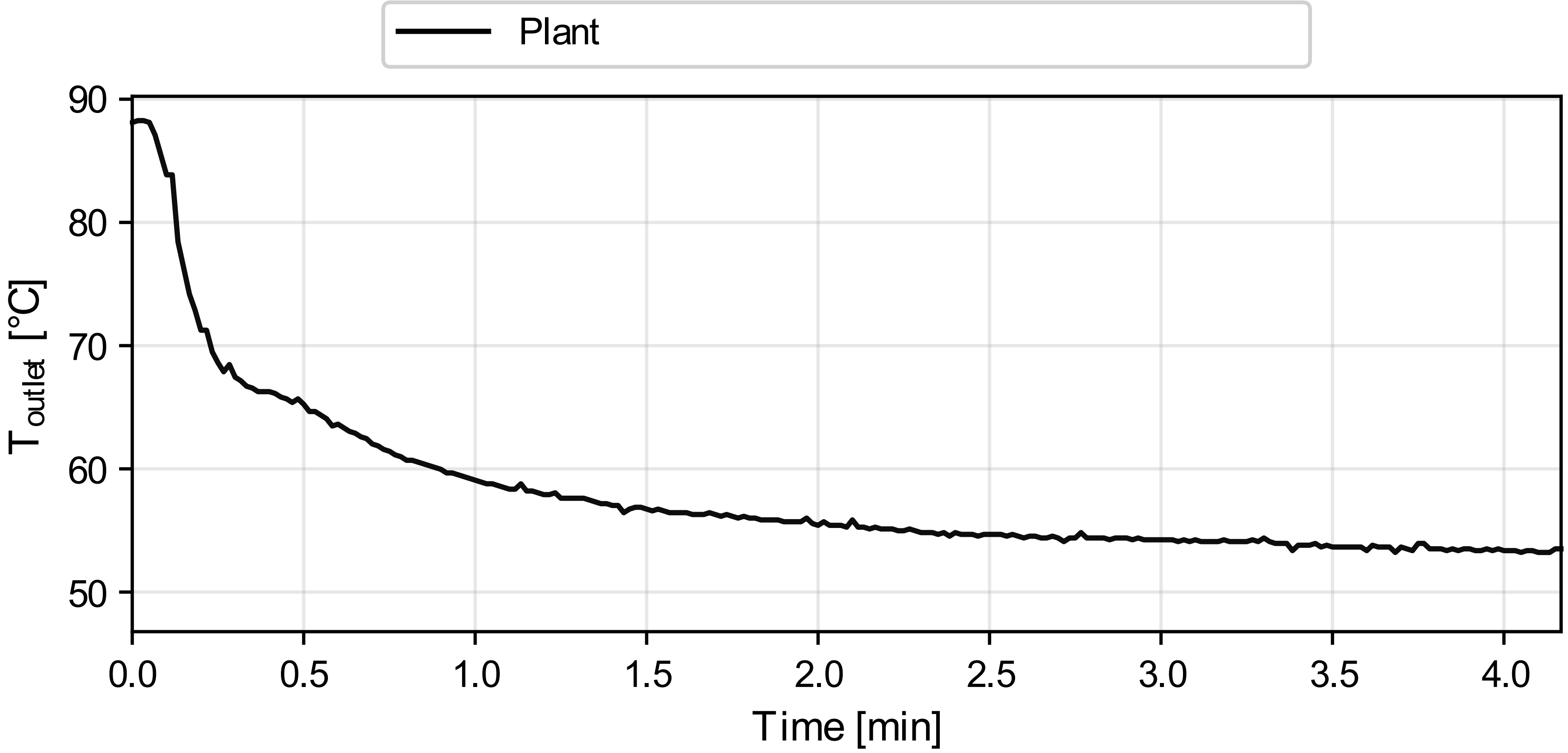
Identification Inputs



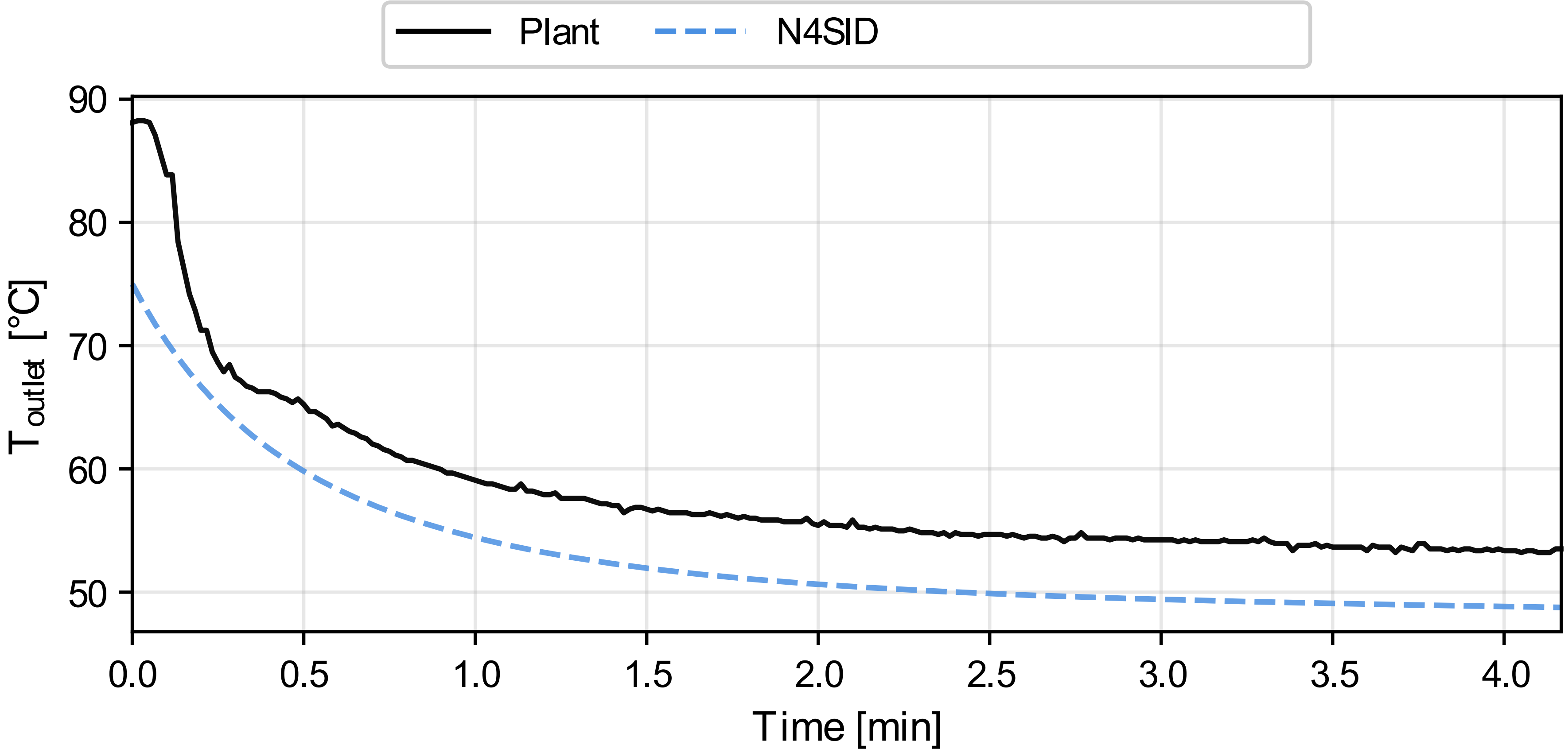
Identification Outputs



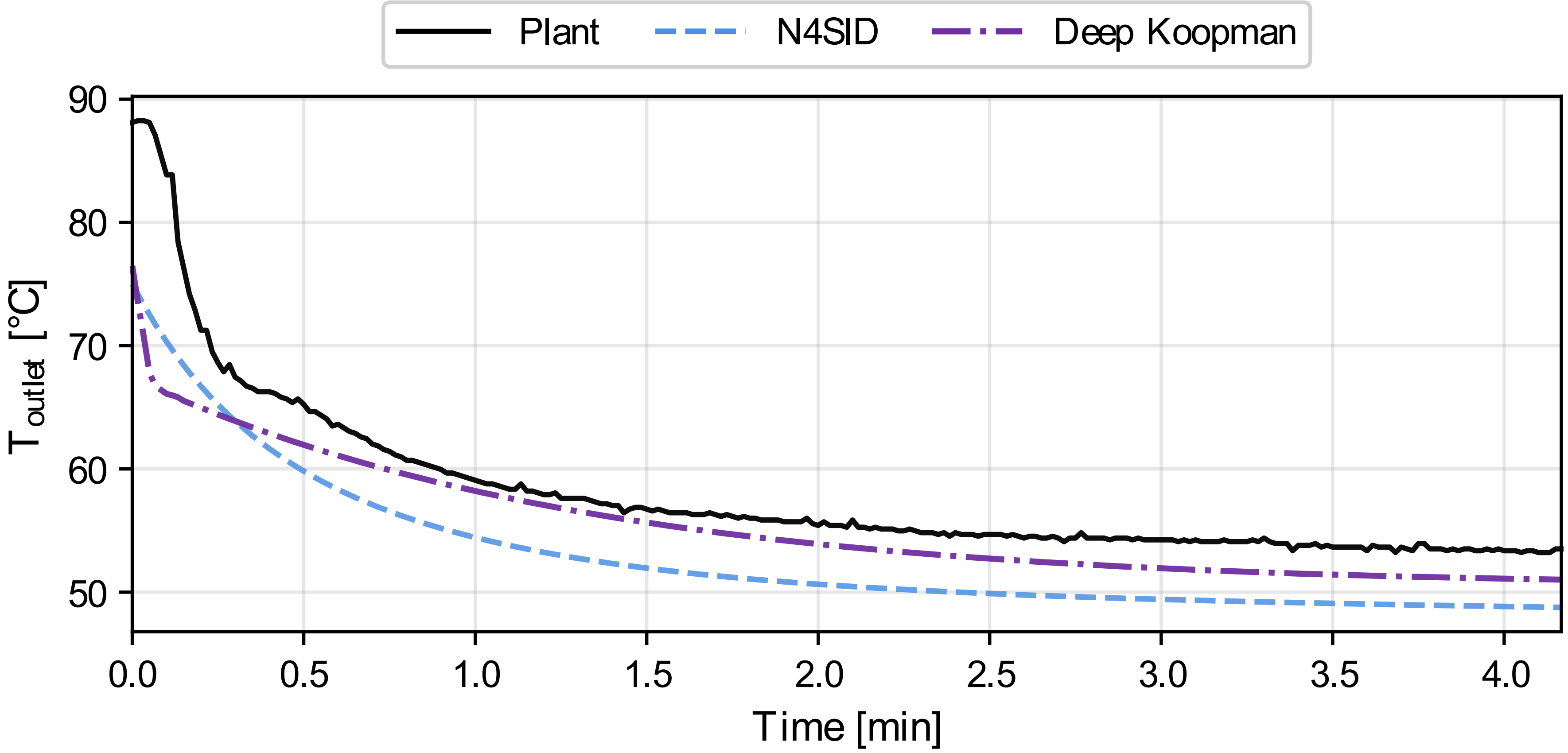
Open-Loop Performance



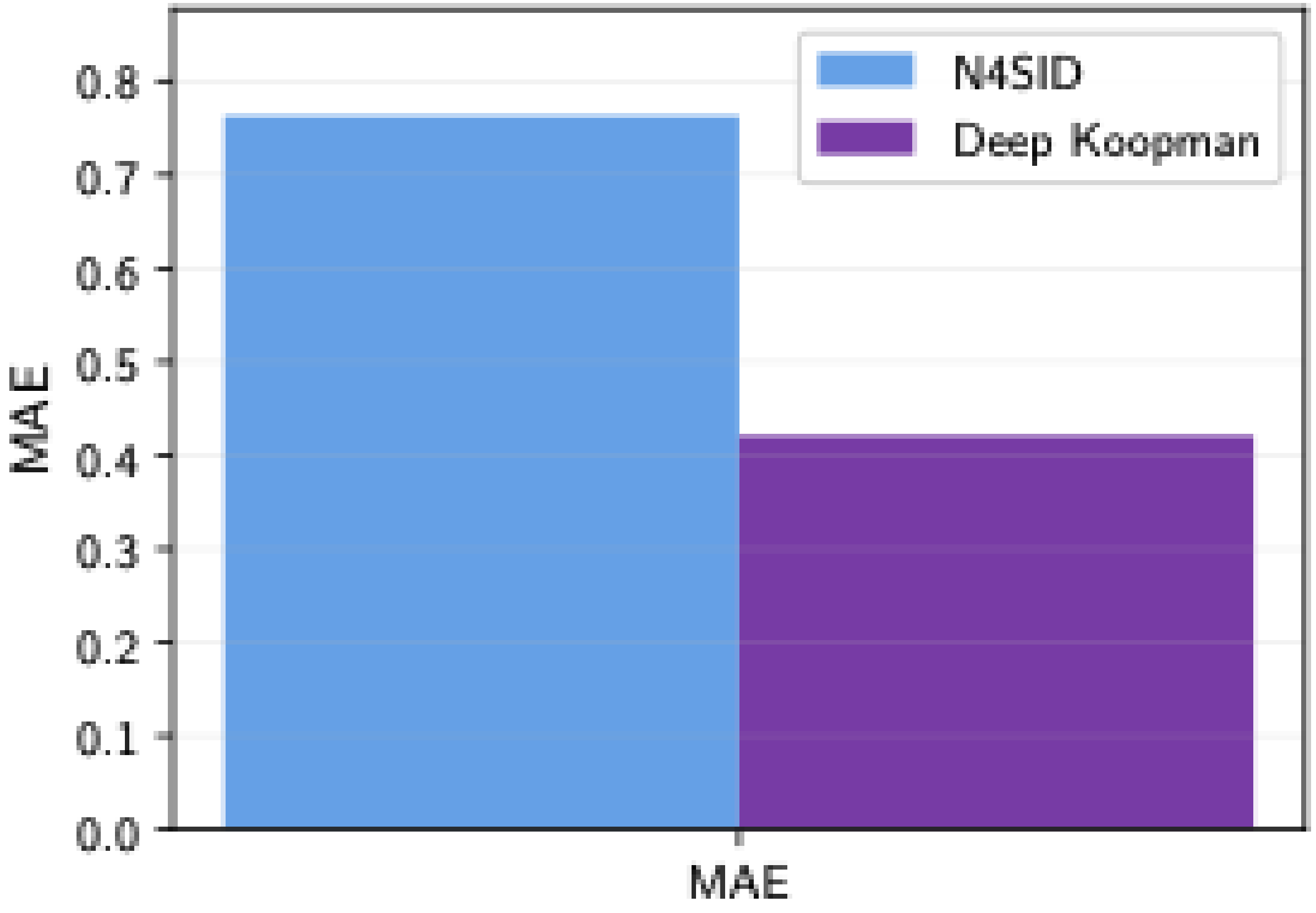
Open-Loop Performance



Open-Loop Performance

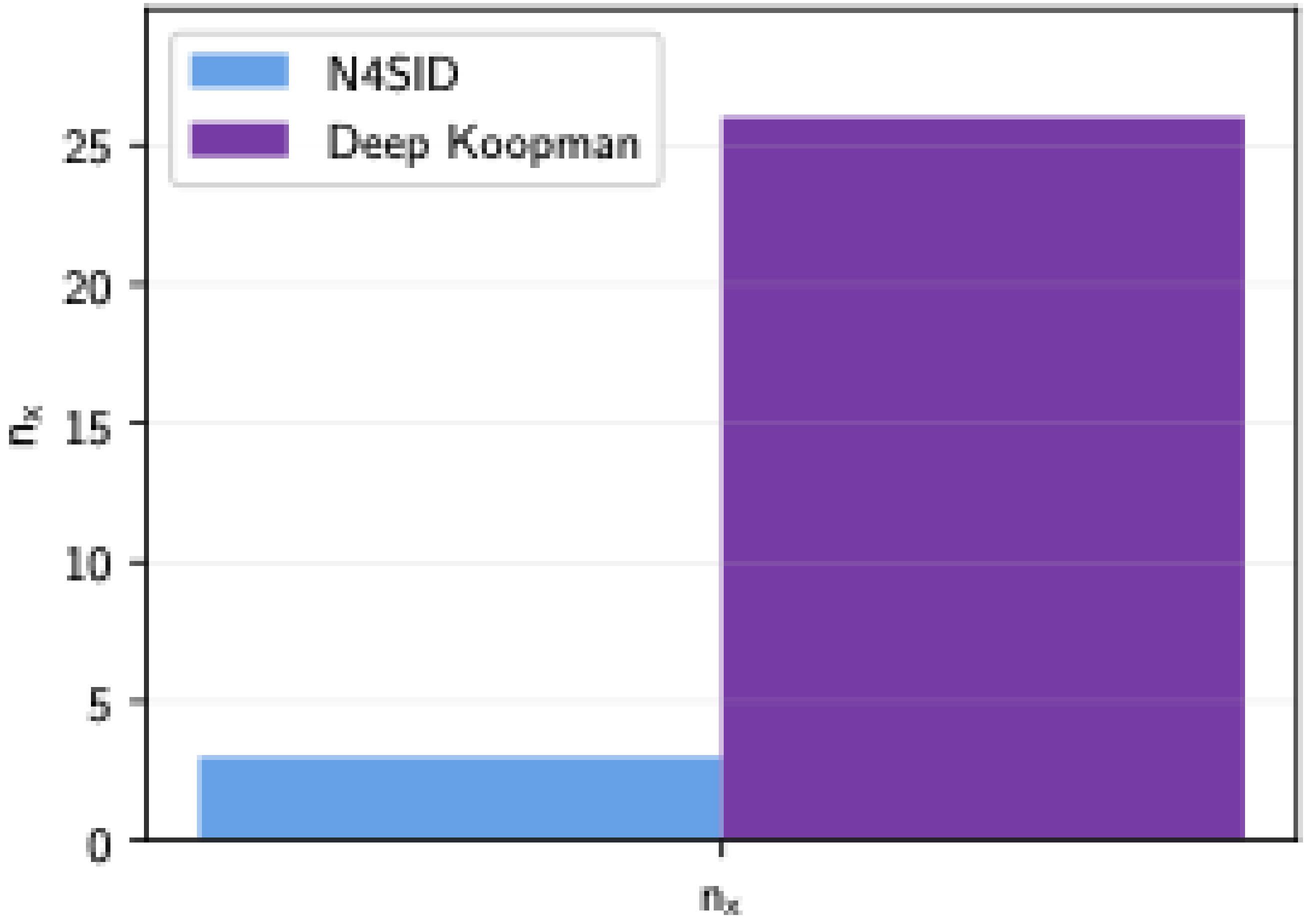


Open-Loop Performance - Numerical

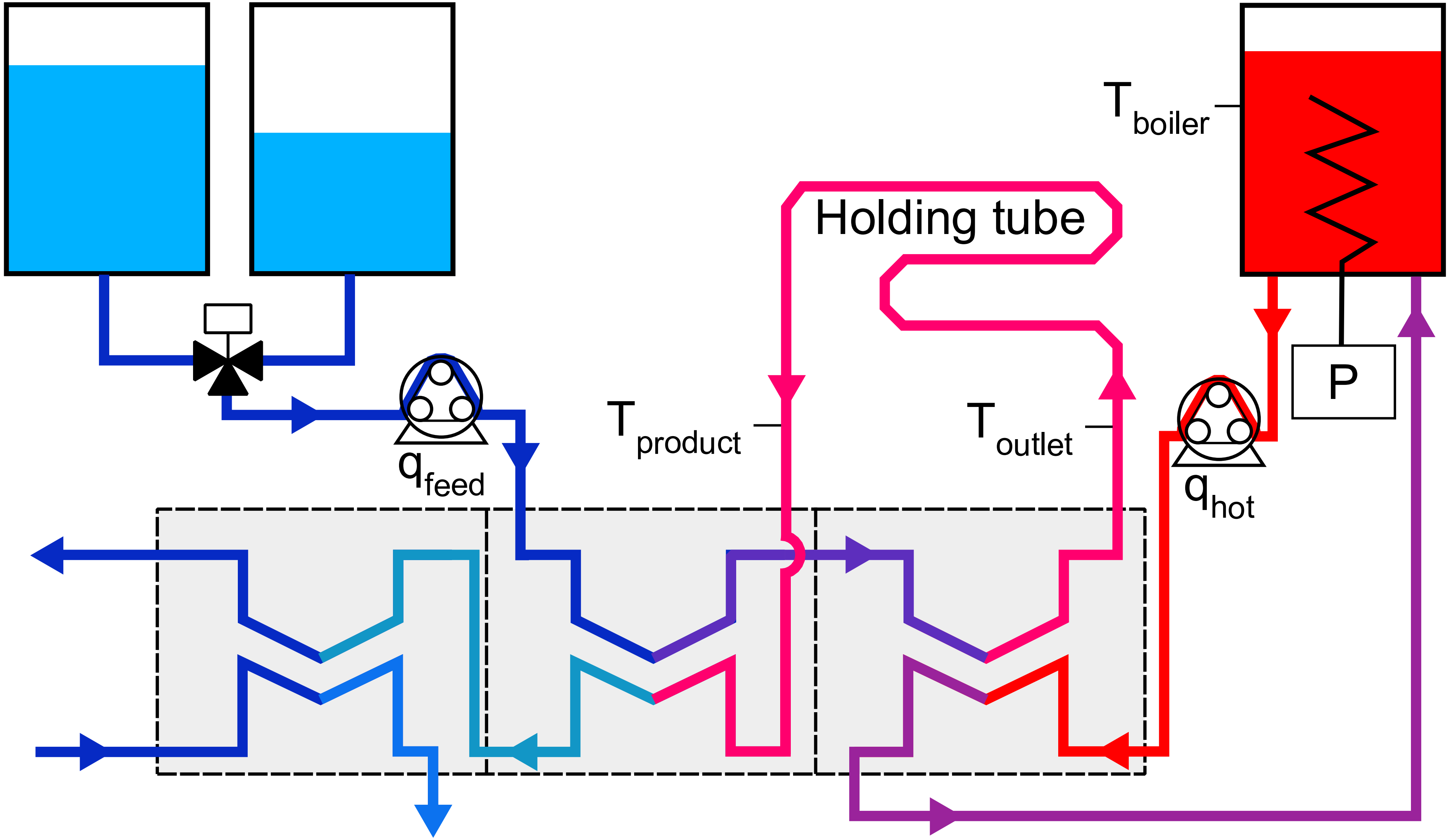


45% MAE improvement

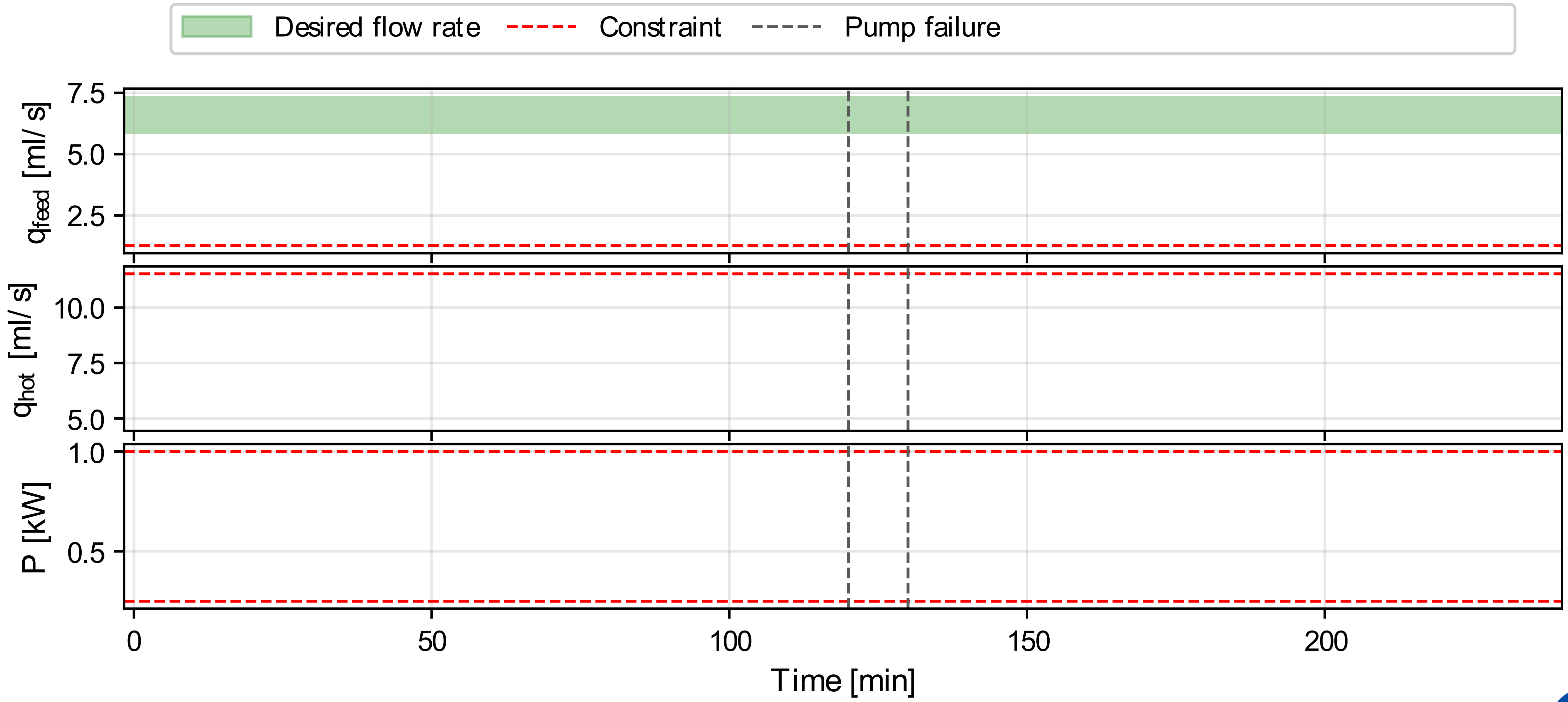
Number of States



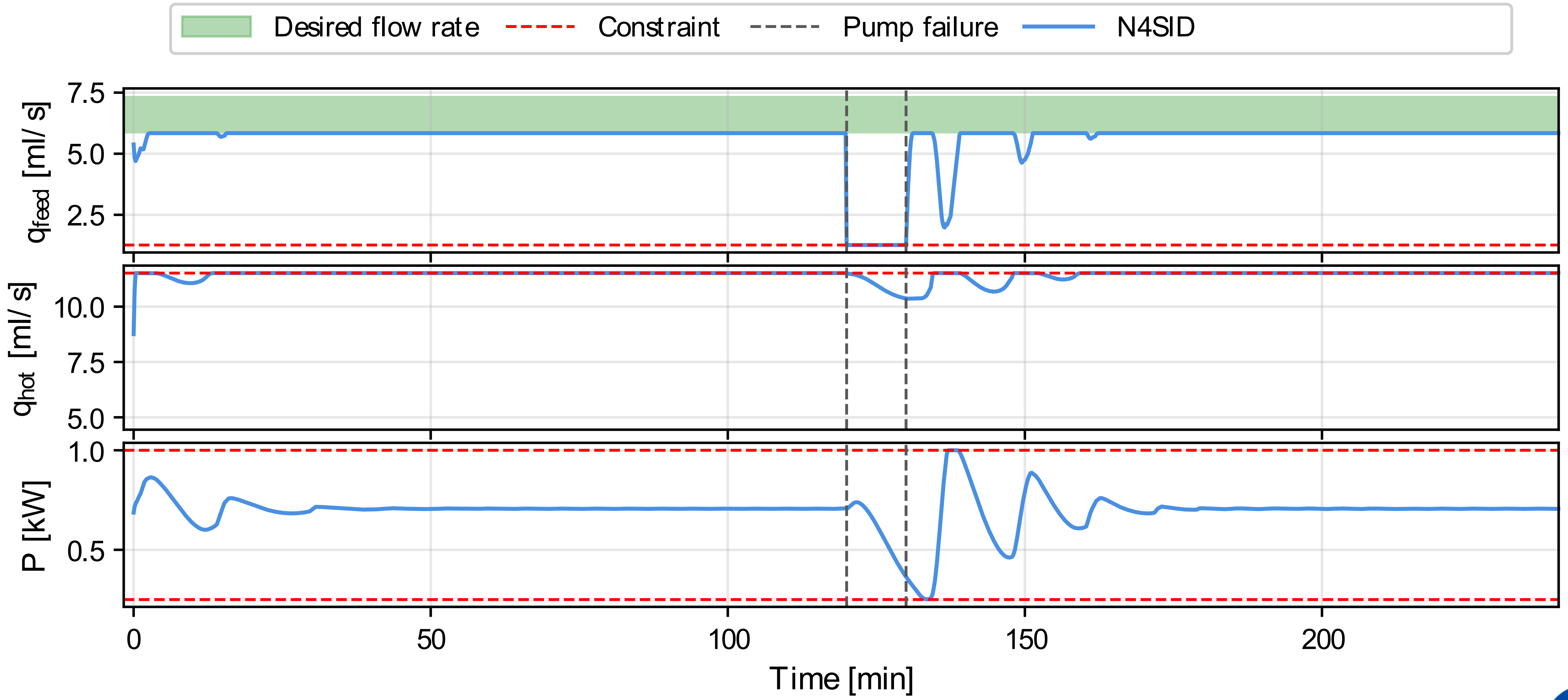
The Laboratory Pasteurization Unit



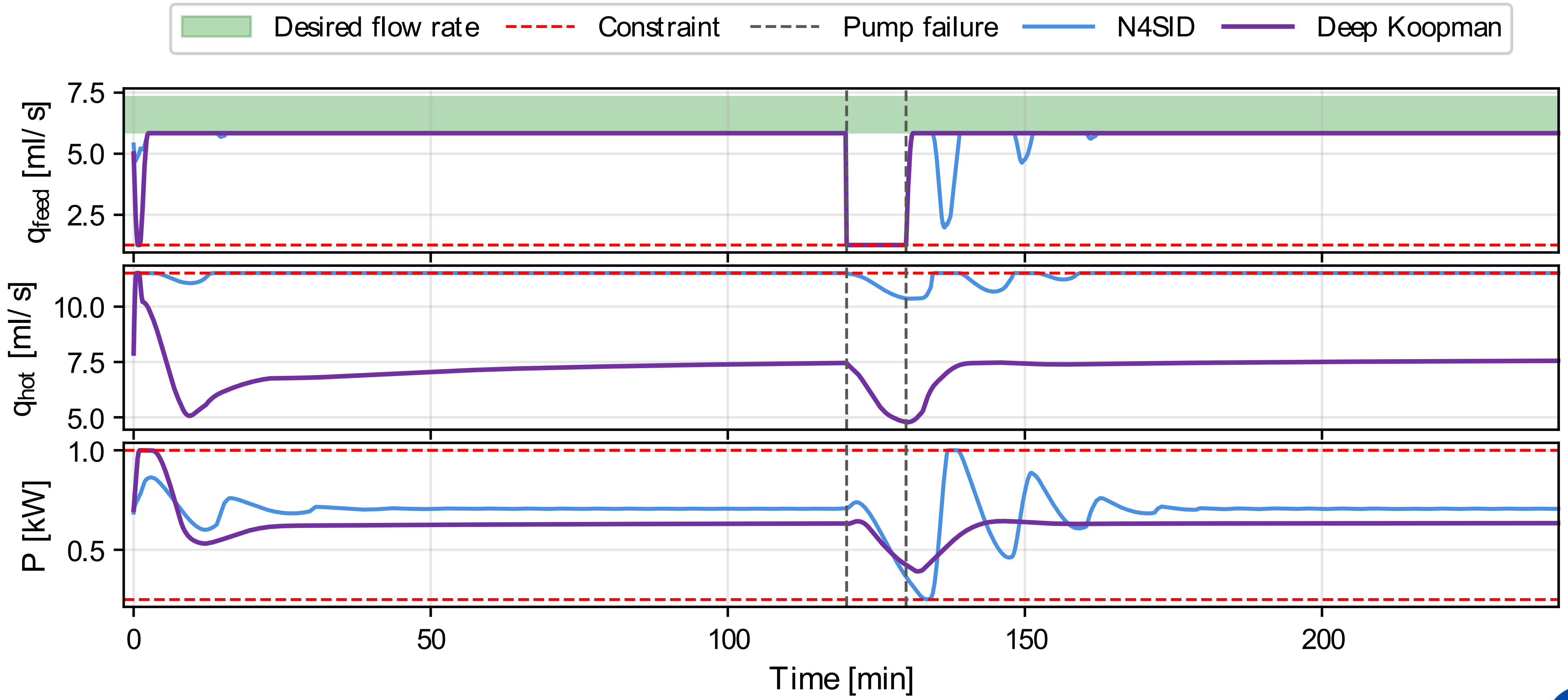
Closed-Loop Inputs



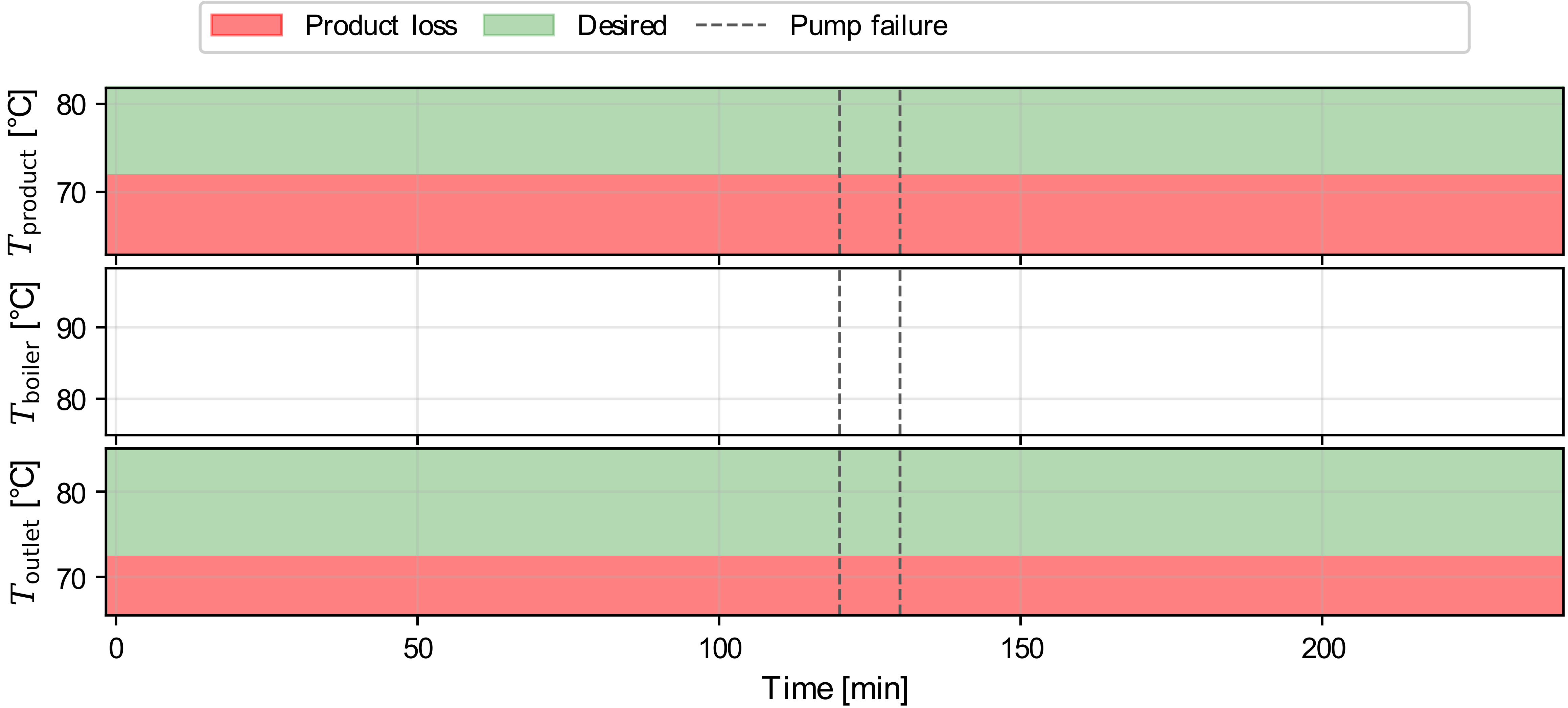
Closed-Loop Inputs



Closed-Loop Inputs



Closed-Loop Outputs



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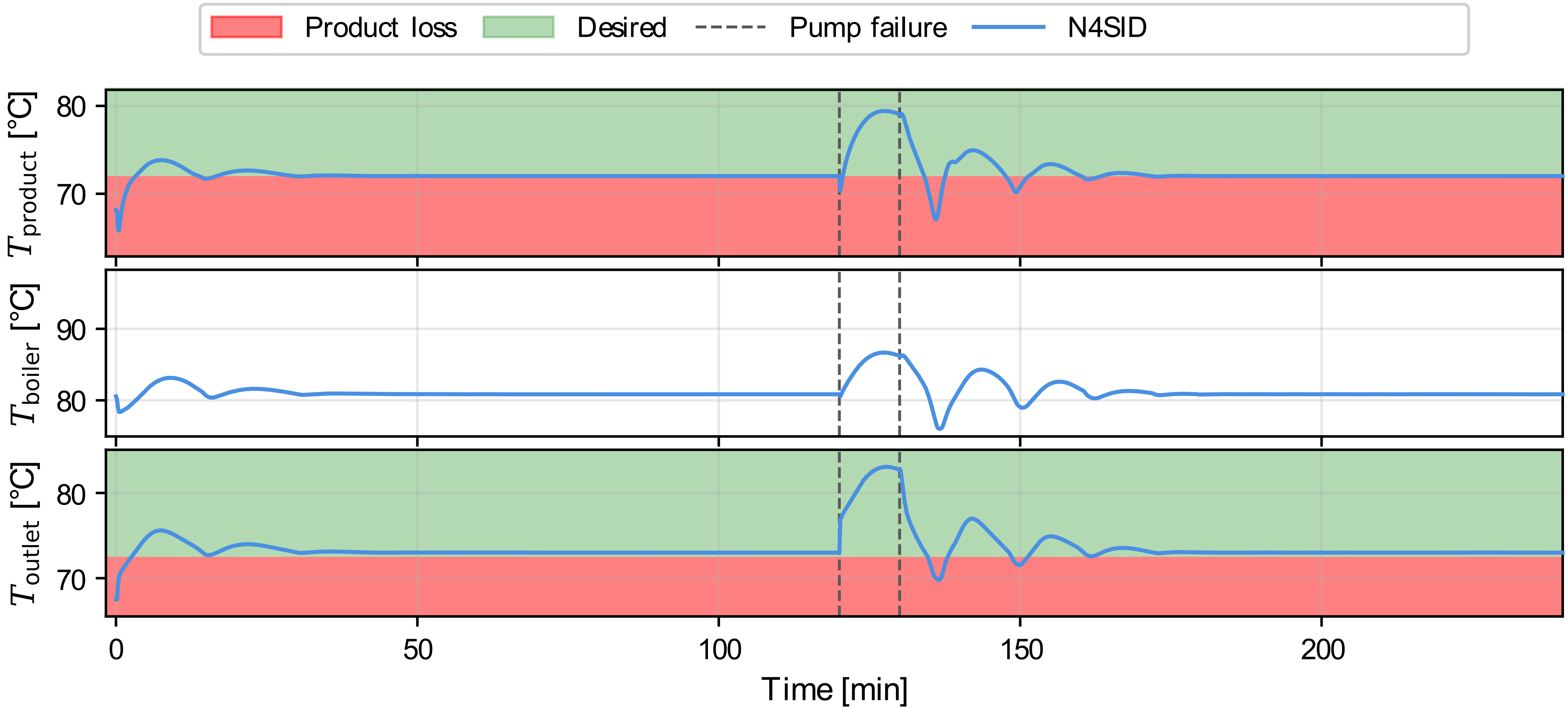

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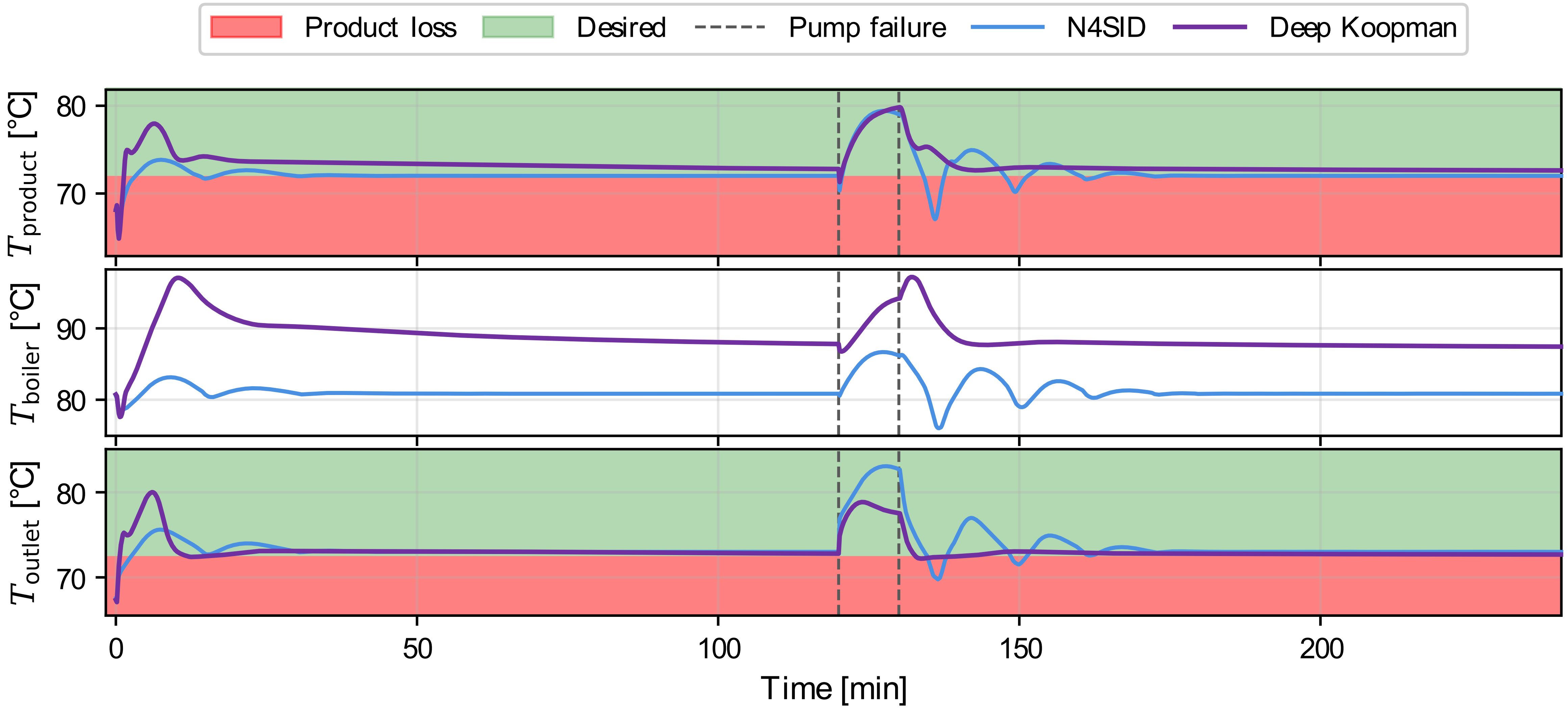
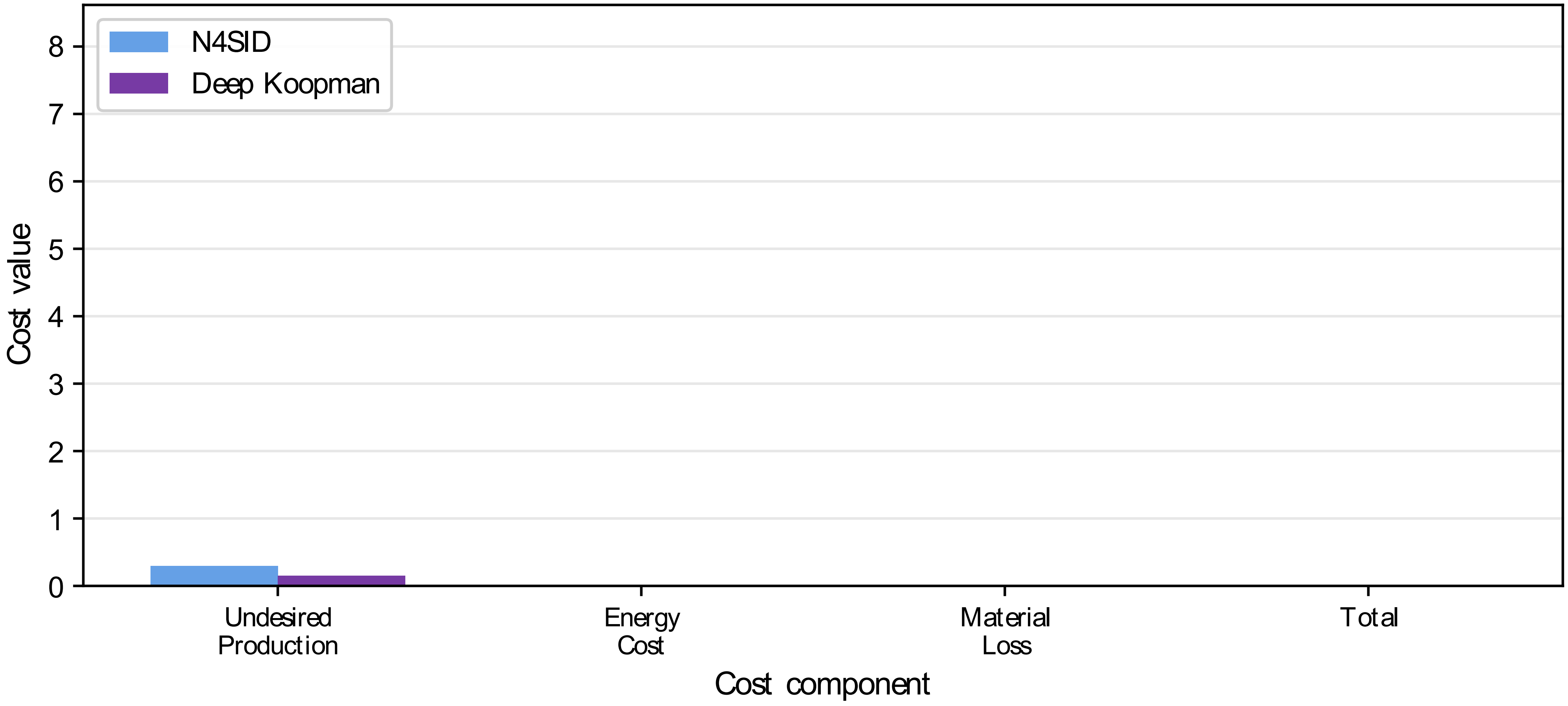


The Economic Evaluation



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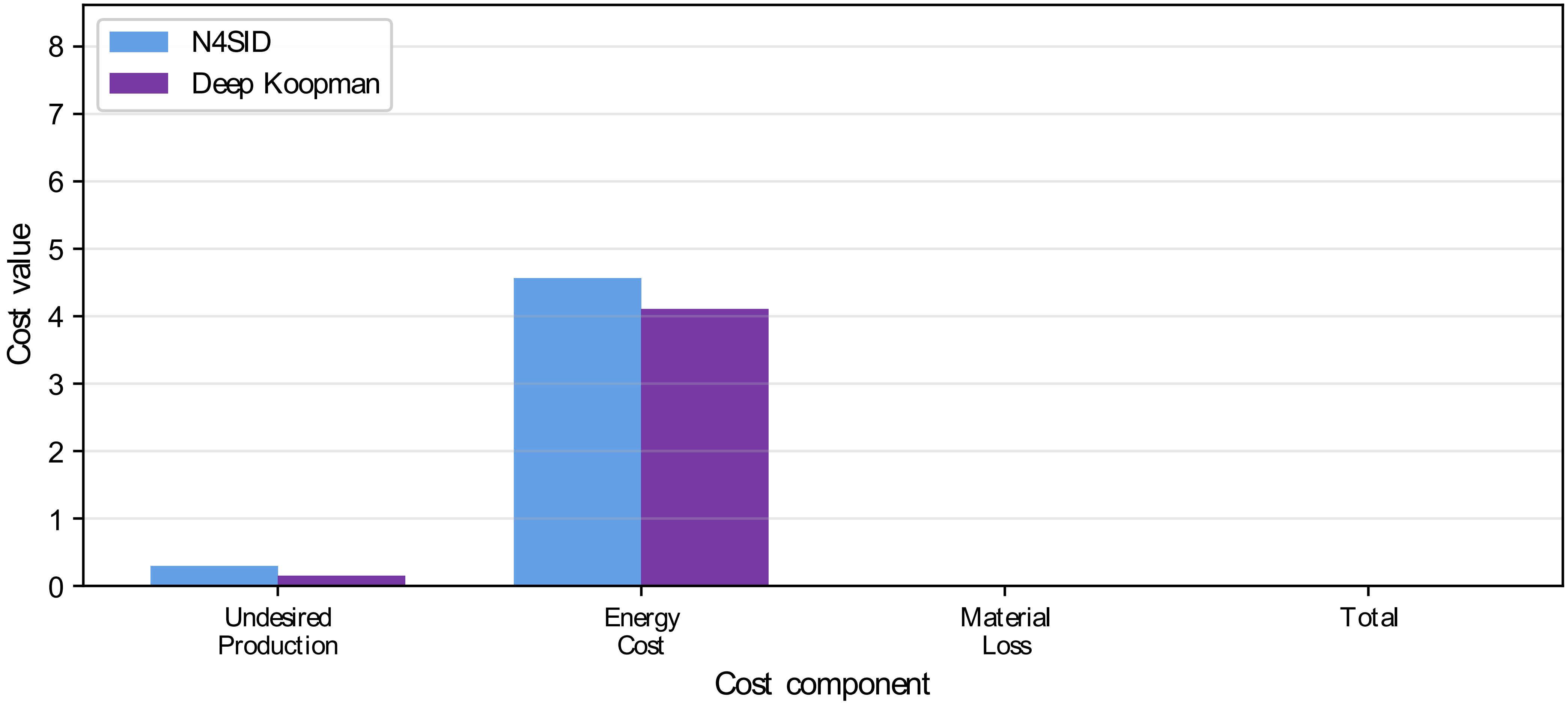

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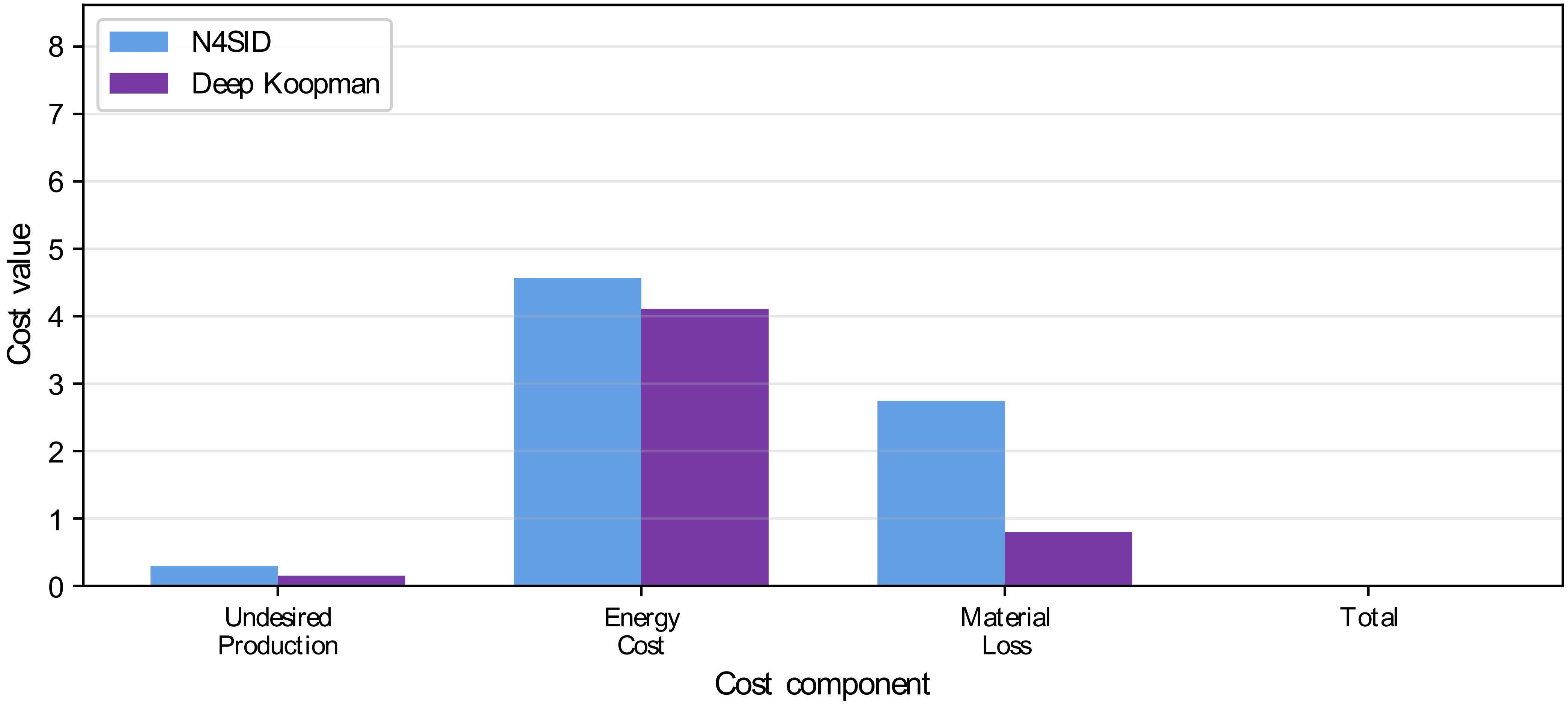

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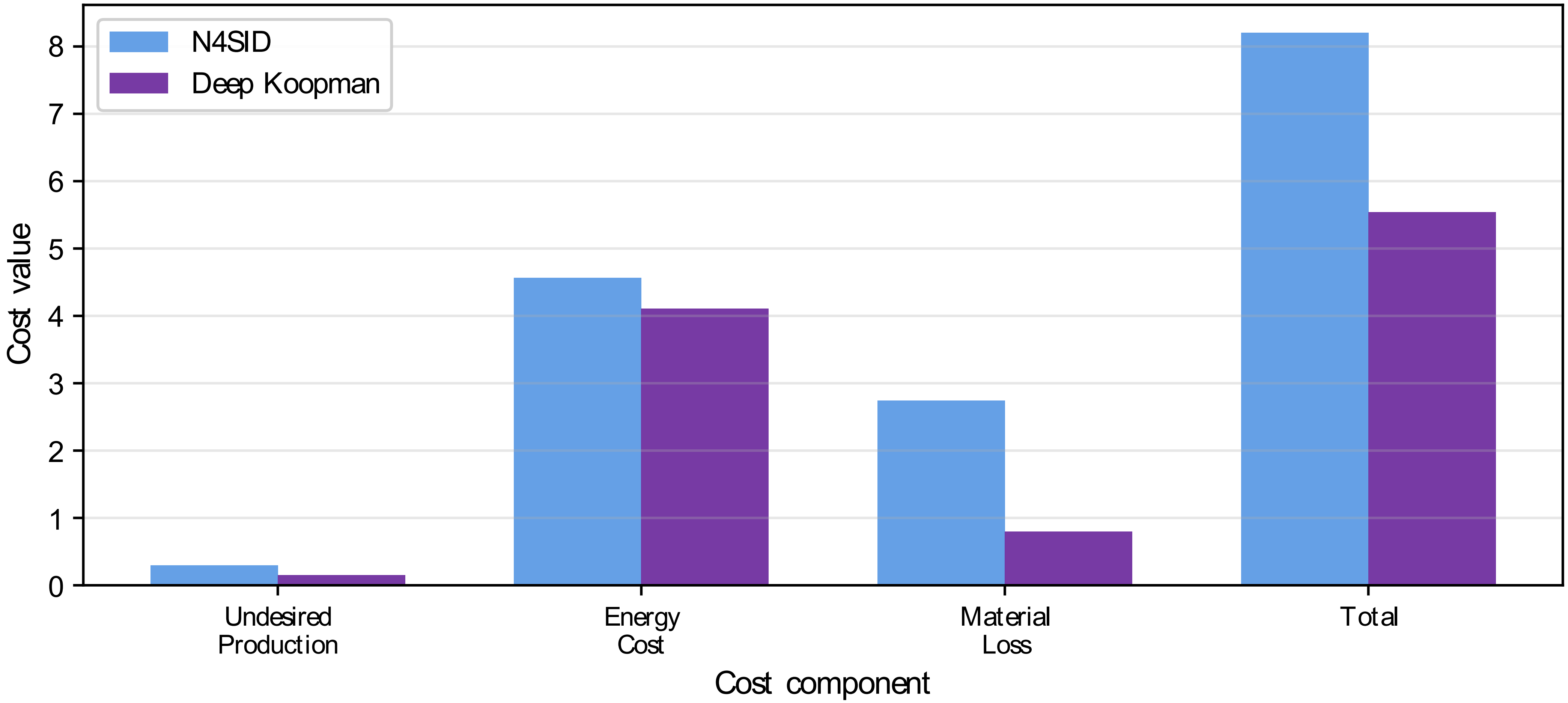

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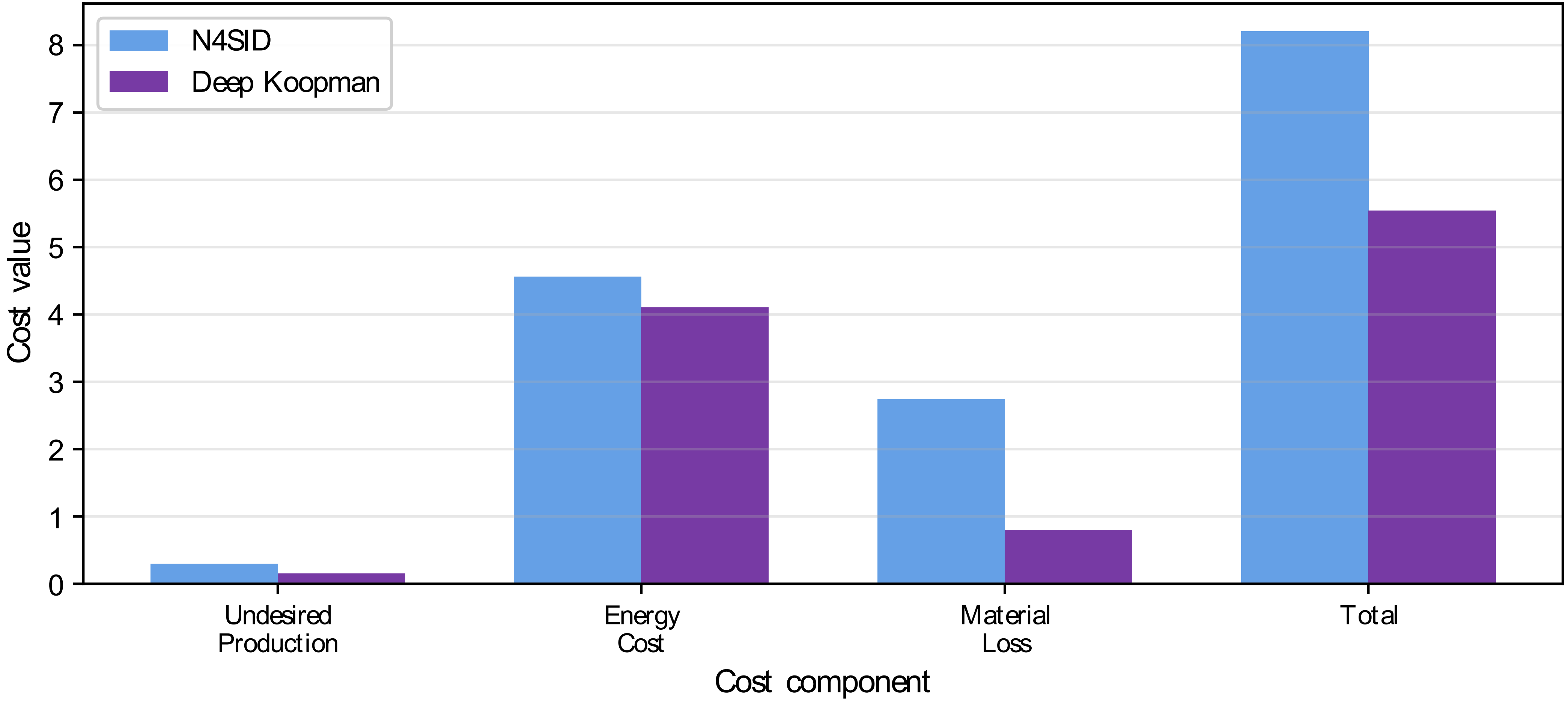

32% of economic savings

Five Questions Answered

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